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PALEOMAGNETISM OF THE MIDDLE SIWALIK FORMATIONS OF NORTHERN PAKISTAN AND ROTATION OF THE SALT RANGE DECOLLEMENT

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ABSTRACT

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Sampling for paleomagnetic study has been carried out at seven localities in the Potwar Plateau region of Pakistan in the vicinity of the Salt Range. In all, 253 sites, consisting of three samples per site, were studied. The sites were taken from the Chinji, Nagri, Dhok Pathan and Upper Siwalik formations which span the last 14 m.y. of Earth history. The formations have been deformed in the ongoing Himalayan orogeny and are exposed today in sections which dip at angles of from 10° to 90° in different directions.

A characteristic direction of magnetization was isolated at all sections by thermal demagnetization at temperatures of from 550° to 650°C. The sediments have been folded within the last 2 m.y. and because of a positive fold test it can be shown that the characteristic magnetization of these sediments was acquired before folding. Four of the sections have directions of magnetization which have been rotated in a counter-clockwise fashion up to 40°; however, sections taken along the Indus River, the north flank of the Soan syncline and the south flank of the Salt Range show small counter-clockwise rotations of under 10°. These observations are in agreement with previous suggestions that strata in the Salt Range have been moved to the south and rotated in a counter-clockwise fashion. The observations reported here indicate that the paleomagnetic poles derived from Cambrian and Permian formations within the Salt Range have probably also suffered rotation and should be removed from consideration in Gondwanaland reconstruction of the Paleozoic.

INTRODUCTION

The collision of the Indian Plate with Eurasia has many spectacular consequences which include the world's highest plateau (Tibet) and the world's highest mountain range (the Himalayan Range). Immediately to the south of these mountains a thick sequence of fluviatile deposits known as the Siwalik Group (Pilgrim, 1910; Pilbeam et al., 1977) was deposited over a period of approximately 15 m.y.

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These sediments are well exposed in the Potwar Plateau of northern Pakistan. The Potwar Plateau is bounded on the south by the Salt Range, on the west by the Indus River, on the north by the Kalla Chita Range and on the east by the Jhelum River (Fig.1). The area includes approximately 20 000 km^2 of Neogene molasse which is covered discontinuously by a thin veneer of Quaternary alluvium, called the Potwar silt. However, large areas of badlands afford excellent exposures locally for stratigraphic and paleontologic study. The Middle Siwalik sediments have been extensively folded into a large system of anticlines and synclines as a result of the continued motion of the Indian Plate and are presently deformed at angles of up to 90°. The present drainage system is being actively incised into these sediments and stream courses, locally called Kas's, provide excellent stratigraphic sections some of which approach 2 km in thickness.

The Siwalik Group has been subdivided into a series of lithostratigraphic units formalized by the Stratigraphic Committee of Pakistan (Fatmi, 1973). The oldest unit is the Kamlial Formation which consists of deep red mudstones and sandstones and may represent a mix of upper delta plain sedimentary

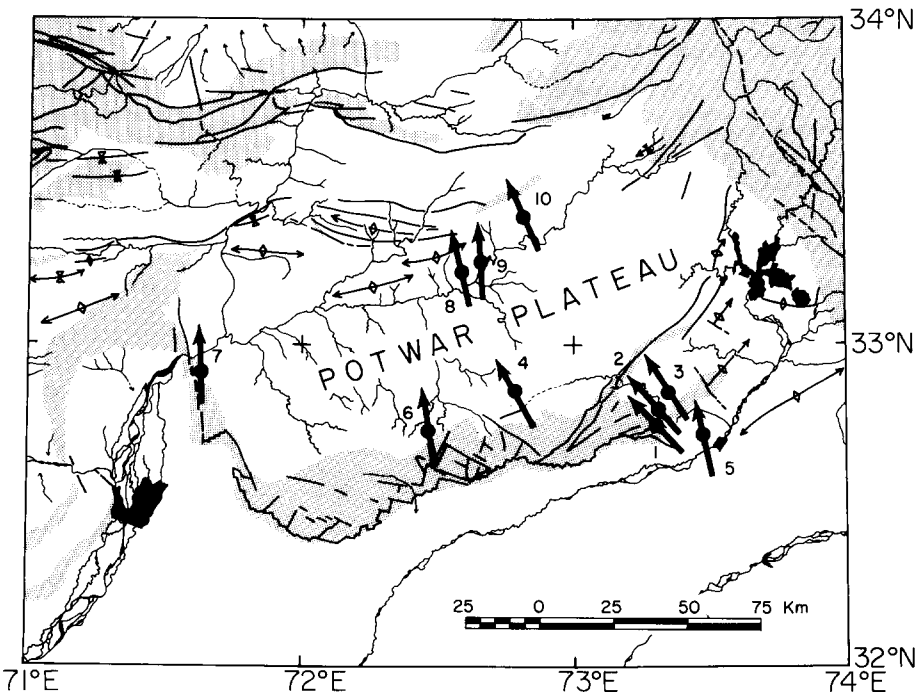


Fig.1. Locality map of sections sampled in the Potwar Plateau region of northern Pakistan: *1* = Kotal Kund, *2* = Dhala Nala, *3* = Tatrot, *4* = Chakwal Bhaun, *5* = Jalalpur, *6* = Chingi-Nagri, *7* = Dhaud Khel, *8* = Hasal Kas, *9* = Rada Kas, *10* = Kamliel Kas. Elevated areas are stippled. Tectonic interpretation after C. Powell (pers. comm.).

facies. This formation was not sampled in the course of the investigation reported here.

Immediately above the Kamlial Formation and transitional with it is the Chinji Formation, the type section of which is defined (Fatmi, 1973) south of Chinji village (72°22'E, 320°41'N) in the southern Potwar. This unit similarly consists of a series of lateral accretion sandstones intercalated with thick red vertical accretion silts. The Nagri Formation succeeds the Chinji Formation in this region conformably. Its type area is in the region 8 km northeast of Chinji village near Sethi Nagri (72°30'E, 32°46'N) (Fatmi, 1973) in the gorge of the Gambir River. The Nagri Formation is composed of a series of thick multi-storied sandstones separated by thin red overbank mudstones and siltstones.

The succeeding Dhok Pathan Formation (Fatmi, 1973) has been defined from the region near Dhok Pathan (72°31'E, 33°8'N) on the northern flank of the Soan syncline. In this area the massive sandstones of the Nagri Formation give way to the thinner sandstones and more massive overbank muds and silts of the Dhok Pathan Formation. The fine-grained deposits of this formation are red in color but not the dark red of the preceding Chinji and Nagri Formations.

In most of the area of the Potwar Plateau the Upper Siwalik brown and drab sandstones and mudstones succeed the Dhok Pathan Formation disconformally. However, in the region both north and south of the Chambal Ridge (e.g., the Kotal Kund section) the Upper Siwalik sediment lies conformably on sediments of the Dhok Pathan Formation. The magnetic stratigraphy for the Upper Siwaliks has been presented by Opdyke et al. (1979).

PALEOMAGNETIC SAMPLING

Three oriented paleomagnetic samples were taken at each site using a Brunton compass. Care was taken to remove weathered sediment by digging shallow pits in order to obtain unweathered samples. Ridge tops which might be exposed to lightning strikes were not sampled. The methods employed are described fully in Lindsay et al. (1982). Sites were taken preferentially from overbank silts, muds or finer-grained clasts. Sandstones were generally not sampled. Seven sections were sampled (see Fig.2 and Table I). Each section consists of 60 or more paleomagnetic sampling sites that span several formations.

LABORATORY PROCEDURES

After the samples were returned from the field, each one was fashioned into a cube nominally 2.5 cm on each side. The samples were then measured on a commercially available spinner magnetometer described by Molyneaux (1971). Almost all samples were easily measurable with NRM intensities ranging between 10^{-7} and 10^{-5} G/cc with the mode located at 10^{-6} G/cc.

Previously reported studies of Siwalik sediments (Wensink, 1972; Keller et al., 1977; Barndt et al., 1978; Opdyke et al., 1979) have relied on partial alternating-field (a.f.) demagnetization to remove secondary components of magnetization. In the initial studies on the sediments reported here, a.f. demagnetization was also attempted. It soon became apparent, however, that in these red colored sediments, a.f. demagnetization was inadequate to remove secondary components of magnetization since in these older formations much of the remanence is carried by hematite (Tauxe, 1979; Tauxe et al., 1980).

The efficacy of partial thermal demagnetization on hematite-bearing redbeds has long been recognized (Irving and Opdyke, 1965). Therefore, progressive thermal demagnetization was applied to representative samples spaced throughout the various sections. Thermal demagnetization was carried out in a Schonstedt Thermal Demagnetizer and in all 113 demagnetization curves were obtained on which the temperature of blanket thermal demagnetization was eventually based.

THERMAL DEMAGNETIZATION

Fig.2a (NG 1C) shows a Zijderveld-diagram of thermal demagnetization for a normally magnetized site from redbeds of the Chinji Formation. In this specimen, which has the magnetization vector directed to the northwest and down, the blocking temperatures are quite "discrete" (Irving and Opdyke, 1965) showing that a considerable proportion of the NRM has blocking temperatures above 655°C. Only a small unstable component is present.

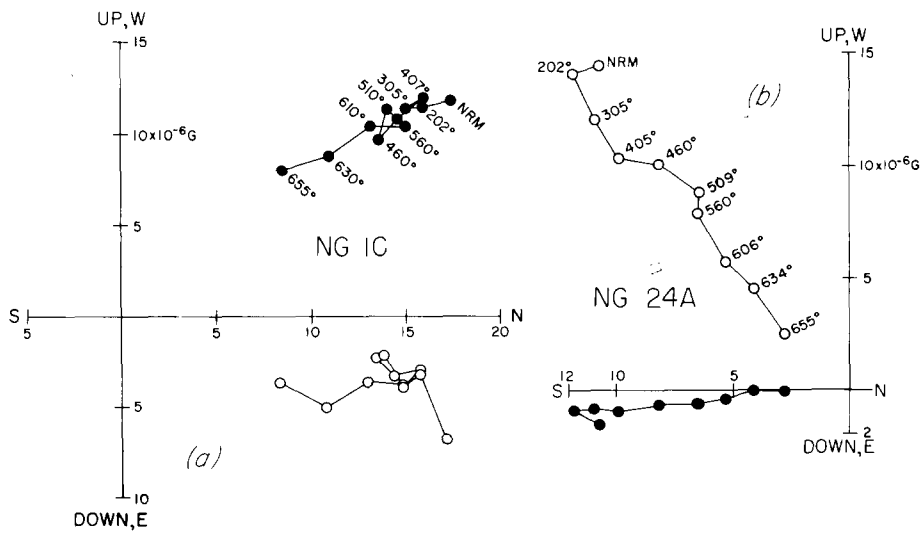


Fig.2. Zijderveld diagram of two samples from the Nagri locality. Closed circles give the horizontal component, open circles the vertical component. Temperature is given in °C.

Fig.2b (NG 24A) shows the results of thermal demagnetization for a reversely magnetized sample from the Nagri Formation. In this sample the characteristic magnetization is distributed through blocking temperatures from 200° to 655°C, and decays linearly to the origin. The small unstable component is removed at low temperatures.

Fig.3 shows Zijderveld-diagrams of the thermal demagnetization for two samples from the Chakwal section (Fig.1), one normal (a) and one reversed (b). Two components of magnetization are present. The low blocking temperature component is in the direction of the present Earth's field. The characteristic direction of magnetization has a positive inclination west of north and decays linearly to the origin with increasing temperature. Fig.3b illustrates the behaviour of a reversely magnetized sample from the Chakwal section. In this case the removal of the low-temperature component causes the growth of the remanent vector. Thus, the unstable component is antiparallel to the characteristic direction of magnetization and is removed by 505°C; above this temperature the characteristic vector decays linearly to the origin.

Fig.4 shows the thermal demagnetization results for two samples from the Tatrot section (Fig.1). TT 67B shows the behavior of a normally magnetized sample. In this case a large unstable component is removed in low temperature; little change occurs over the temperature range 250—450°C. Then the vector increases and a high-temperature component subsequently decays to the origin. The difference in direction of the two components is slight but

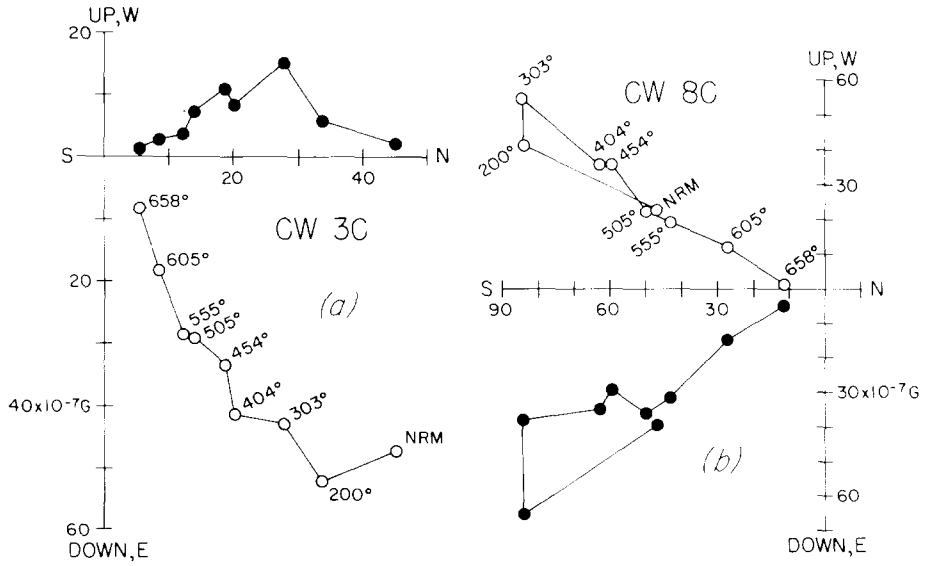


Fig.3. Zijderveld diagram of two samples from Chakwal section. The symbols are as given in Fig.2.

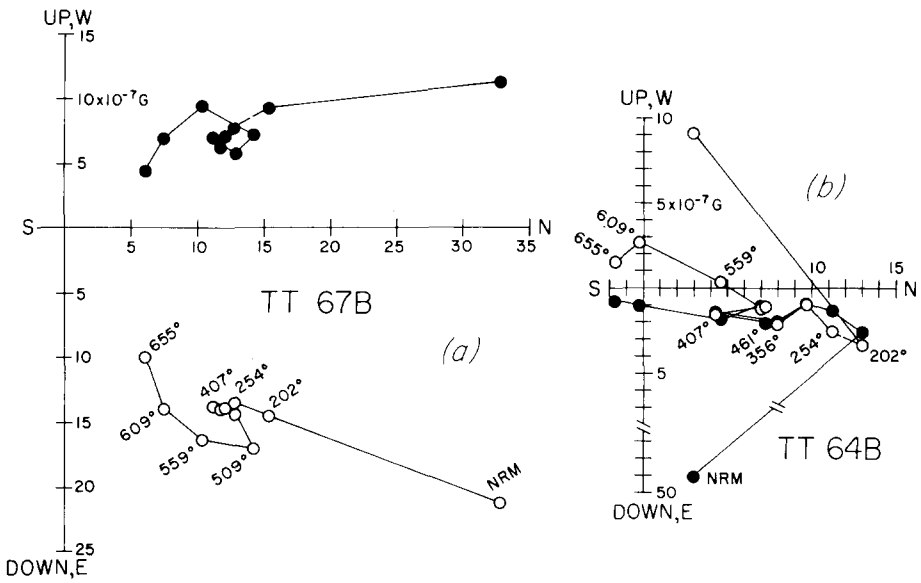


Fig.4. Zijderveld diagram of two samples from the Tatrot section. The symbols are as given in Fig.2.

significant. The behavior of TT 64B (in b) is in many ways similar but in many respects more striking. Again, a large very low coercivity component is removed at temperatures below 200° C. A harder component which is aligned about the present field is then removed at temperatures between 200 and 550° C. However, at temperatures exceeding 600° C, the declination moves into the southeast quadrant and the inclination becomes negative. In this sample the characteristic magnetization persists at temperatures exceeding 600° C indicating that the magnetization is probably carried by hematite.

Immediately to the southwest of the Tatrot section another section was sampled along Dhala Nala. A sample normal (a) and reversed (b) from this section are shown in Fig.5b. These samples exhibit a steady uncomplicated decay to the origin and most contain only a single component from 200° C to the Neel point of hematite. The polarity and direction are unambiguous in this case.

To the south of Dhok Nala and north of the Chambal Ridge, an important section was taken in the Kotal Kund region (Fig.1). Fig.6 illustrates the behavior of two samples from this section. The sample in (a) (KK 144B) is normally magnetized with at least two components: the weaker component is removed at 300° C; the magnetization vector of the stronger component remains constant to destruction. The behavior of the other sample (b) is similar. In this case the stronger component is reversely magnetized whereas the large overpring is normal and is removed again at 400° C; the direction of magnetization remains fairly stable up to 625° C where most of the magnetization is lost.

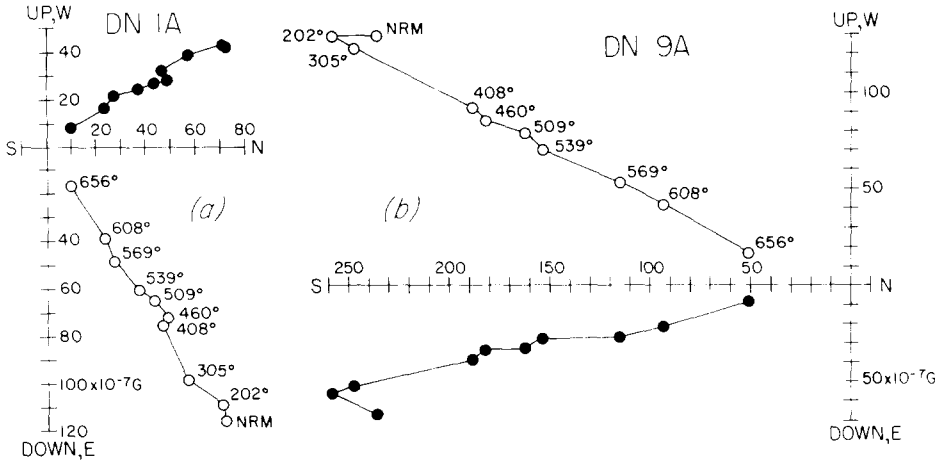


Fig.5. Zijderveld diagram of two samples from the Dhala Nala section. The symbols are as given in Fig.2.

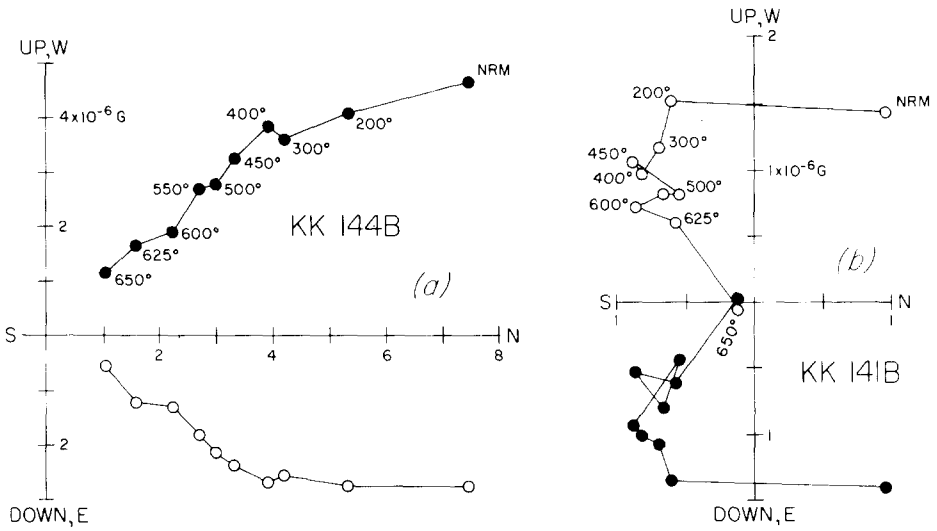


Fig.6. Zijderveld diagram of two samples from the Jalalpur section. The symbols are as given in Fig.2.

As a result of this extensive demagnetization program, temperatures were chosen for blanket demagnetization at one temperature. The resulting directions of magnetization are given in Table I with the locality mean statistics. If the results from the three samples collected at each site were statistically significant on the Watson and Irving test (1957), the results were combined into locality means directions.

TABLE I

Locality mean directions

Locality	Lat. (°N)	Long. (°E)	Normally magnetized sites					Reversely magnetized sites					Combined sites				
			N	D	I	α_{95}	k	N	D	I	α_{95}	k	$\hat{N}$	$\hat{D}$	$\hat{I}$	$\hat{\alpha}_{95}$	$\hat{k}$
Kotal Kund	32.75	73.30	37	323.4	30.6	5.8	17.3	29	137.0	−38.5	7.2	14.9	66	320.8	34.1	4.6	15.6
Dhala Nala	32.85	73.20	14	329.4	36.3	6.9	33.9	5	146.0	−34.7	11.6	44.5	19	328.5	35.9	5.5	37.6
Tatrot	32.85	73.35	45	325.6	36.7	4.4	23.9	36	150.5	−36.7	6.9	12.8	81	327.7	36.7	3.9	17.3
Chakwal Bhaun	32.85	72.75	25	328.3	33.7	8.2	13.4	16	154.2	−28.6	10.8	12.6	41	330.7	31.8	6.4	13.1
Chambal Ridge	32.70	73.45	49	355.2	33.0	4.5	21.6	38	166.3	−43.1	5.8	17.2	87	351.7	37.4	3.7	17.8
Nagri	32.75	72.45	26	347.8	36.2	7.0	17.6	16	175.8	−32.3	8.3	20.8	42	350.9	34.8	5.3	18.4
Dhaud Khel	32.90	71.65	4	356.3	37.2	14.1	43.2	4	176.8	−34.6	16.5	32.0	8	356.6	35.9	8.6	42.4

FOLD TEST

Fig.7 shows the distribution of the mean directions of the Class 1 sites for all the localities reported in this paper. It can be seen that several of the localities have directions which cluster off the north—south axis. Some, however, do not. In order to test for stability, three of the localities which were sampled on different limbs of the Bunha syncline were examined to see if the stable directions had been acquired before folding of the beds took place. Fig.8 shows the directions before folding (circles) and after folding (triangles). By correcting for the tilt of the beds, the mean directions for these three localities (Kotal Kind, Dhala Nala and Tatrot) become very tightly grouped. The decrease in dispersion is significant at the 95% level according to the test described by McElhinny (1966). This is strong evidence that the directions of magnetization isolated by thermal demagnetization were acquired before folding. At all localities normal and reversely magnetized sites occur. It can be seen from an examination of Table I that the reversed and normal magnetized sites are antiparallel with respect to each other, thus successfully passing the reversal test which suggests that the characteristic directions of magnetization were acquired early in the history of the sediment.

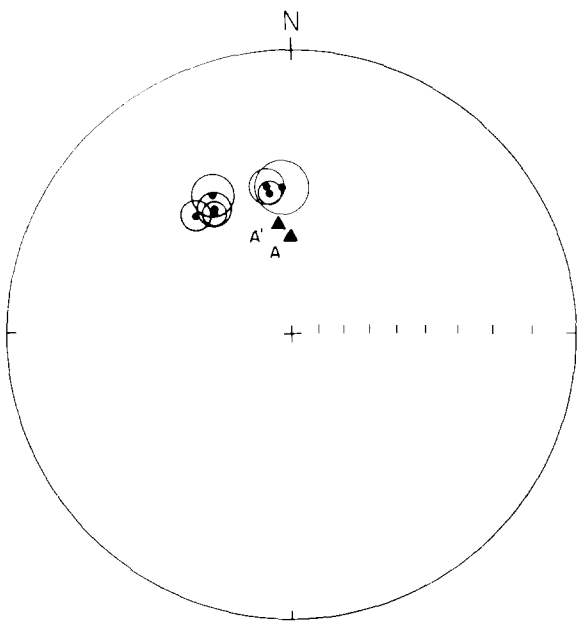


Fig.7. Locality mean directions with circles of 95% confidence. Filled triangle *A* indicates the direction of the present dipole field. *A'* gives the direction of the dipole field at 10 m.y.B.P. determined for the region after being translated and rotated by the amount indicated by the displacement of the Indian Plate as given by Molnar and Tapponier (1976). The points are plotted on an equal angular stereographic projection.

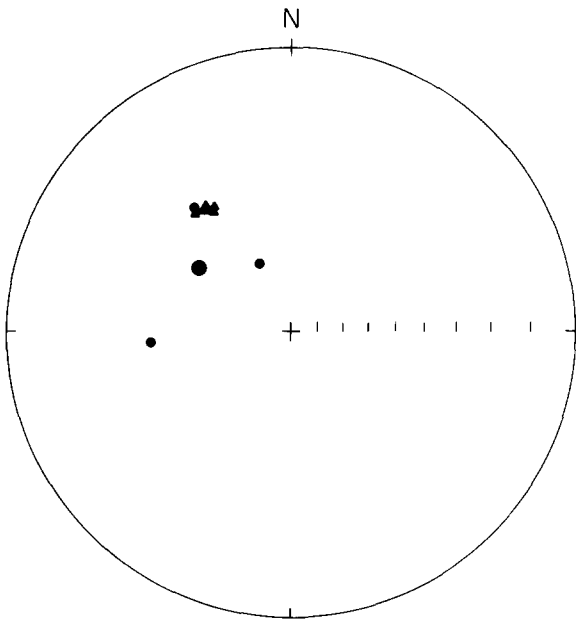


Fig.8. The locality mean direction from the Bunha River region plotted on a stereographic projection. The filled circles are the locality means uncorrected for bedding tilt. The filled triangles are the locality mean directions after correction for the tilt of the beds. The large symbols are the grand means of the three localities before and after rotation.

In a companion study of the Khaur anticline area, Tauxe et al. (1980) have shown that in sediments correlative with those reported here, the characteristic magnetization resides in the high temperature component (above 600°C) and is carried by hematite. The sediments successfully pass a conglomerate test leading the authors of that study to conclude that the magnetization was acquired early in the history of the sediments.

From the present study and from Tauxe et al. we can conclude that: (1) the characteristic magnetization is carried by hematite and in some cases by both magnetite and hematite; (2) the magnetization is older than the deformation of these sediments which postdates the youngest sediments sampled which are approximately 1.5 m.y. in age (Opdyke et al., 1978; N. M. Johnson et al., 1982); (3) the magnetization is probably acquired early, probably before lithification during deposition or during soil formation in the overback areas; and (4) since the sedimentary cycles (sandstone-mudstone-couplets) can be shown to be about 50 000 years in length (Opdyke et al., 1979; Barndt et al., 1978) it is most probable that the characteristic magnetization is acquired within 50 000 years of deposition. The sediments are therefore excellent for use in magnetostratigraphic studies.

ROTATION OF PORTIONS OF THE SALT RANGE DECOLLEMENT

Fig.7 gives the locality mean directions of magnetization after correction for bedding tilt along with their respective circles of 95% confidence. It can be seen that the localities can be separated into two sets based on the mean directions of magnetization. The first group represents results from three localities, Chambal Ridge, Nagri, and Dhaud Khel (Fig.1), and has directions which cluster west of north with declinations ranging between 350° and 360° and inclinations ranging between 34° and 37°. The second group from Chakwal, Bhaun, Tatrot, Dhala Nala and Kotal Kund has mean vector directions which group between 320° and 330° with inclinations ranging from 32° to 37°. Both groups are tightly clustered with overlapping circles of confidence. It is clear from these results that the second group has been rotated in a counter-clockwise fashion by about 35° since deposition and before folding as shown by the positive fold test (Fig.8).

The first group does not show much rotation. However, it does give mean directions of magnetization which are systematically to the west of north. Molnar and Tapponnier (1975) have suggested, based on an analysis of the northward motion of the Indian Plate derived from sea floor-spreading anomalies, that the Indian Plate has undergone an anti-clockwise rotation of 6.5° in the last 10 m.y. The sediments studied here range in age from 12 m.y. to 2 m.y. and should, therefore, record this rotation. It seems reasonable to suppose that the bias toward westerly declinations seen in the first group data reflects this rotation since it is about the amount expected. However, the larger rotations seen in the second group cannot be accounted for by the rotation of the Indian Plate. In Fig.1, the mean declinations for each locality are plotted together with the main structural elements of the Potwar region of northern Pakistan. Also plotted on Fig.1 are the mean declinations from three localities on the Khaur anticline which are taken from the study by Tauxe and Opdyke (1982). The area which exhibits the largest rotation is in the southeast and eastern region of the Potwar Plateau. The only locality included in this study which is south of the Salt Range proper is the Jalalpur locality which was sampled along Jamarghal Kas. The rotation here is very slight; however, immediately to the north of the Chambal Ridge three localities show large counter-clockwise rotation. These localities — Kotal Kund, Dhala Nhala and Tatrot — are in a region separated from the rest of the Potwar Plateau by the Bakrala Ridge. About 30 km west of the Bakrala Ridge, the Chakwal-Bhaun Locality also shows significant counter-clockwise rotation. Furthest west at Dhaud Khel along the Indus River no significant rotation is observed. To the north the two localities on the south flank of the Khaur anticline show little rotation. However, the easternmost locality situated along Kulial Kas does show some rotation.

The regional pattern indicates that the observed rotations do not pertain to the entire Indian Plate but are confined to portions of the east-central Potwar Plateau. The rotated area of the Bunha syncline is separated from the

main part of the Potwar Plateau by the Bakrala hills. If the rotation had been confined to this area, this could be ascribed to a local thin-skinned effect. However, the rotation in the Bhaun area is as significant as that further to the east. To the west, at Nagri, little rotation is observed. Complicating the situation is the lack of any significant structural element between the two areas. The relationship of the sections on the flank of the Salt Range with those near the Khaur anticline (to the north) is obscured by the Potwar silt which covers the axis of the Soan synclinorium, and obscures intervening structures. Further to the east, in the Rhotes anticline, a recent paleomagnetic study by Opdyke et al. (1979) in sediments spanning the last 5 m.y. of Late Neogene time gives no indication of significant rotation.

Several authors (e.g. Seeber and Jacob, 1977) have suggested that the Salt Range has resulted from the southward movement of a thrust sheet on a decollement surface of Cambrian salt. Such an explanation of the regional structure would seem to be compatible with the results we have obtained. A plausible model is passive rafting to the south of a portion of the Siwaliks depositional basin with its Upper Paleozoic basement. This is accompanied by variable counter-clockwise rotations along the leading edge of the decollement. This motion would produce the folding of the sedimentary basins into the observed structures.

TIMING OF THE DEFORMATION

South and east of the Potwar Plateau, Opdyke et al. (1979) have shown that deformation has continued in the area and affects sedimentary sequences of early Brunhes age. In all likelihood this deformation is still proceeding. In the sections considered here, four contain Upper Siwalik sediments younger than 5 m.y. based on magnetic stratigraphy (N. M. Johnson et al., 1982) and fission track dating (G. D. Johnson et al., 1982). The youngest sediments in the sections considered here occur in the Jalalpur section, where sediments as young as Middle Matuyama Chron have suffered tilting but little rotation (G.D. Johnson et al., 1979; Opdyke et al., 1979). North of the Chambal Ridge, at Kotal Kund, sediments as young as earliest Matuyama have been tilted and rotated. Further to the north, at the type locality of the Tatrot fauna which overlies the Dhok Pathan Formation, the Dhok Pathan beds are rotated and although the number of sites are few, the Upper Siwalik also appears to be rotated. We had previously correlated these beds tentatively to the Gauss chron.

At the Chakwal Bhaun section, the Upper Siwalik sediments overlie the Dhok Pathan Formation with the base of a coarse conglomeratic sandstone. The Upper Siwaliks are both reversely and normally magnetized and yield a mean declination not significantly different from the north–south axis. These results indicate that rotation of this area took place before deposition of the Upper Siwalik beds, which were tentatively correlated with the Gilbert chron by Opdyke et al. (1979).

The evidence, therefore, favors rotation of the Bhaun sedimentary basin after the onset of Upper Siwalik deposition but before the deformation of the Upper Siwalik beds, between 2 and 2.5 m.y. ago within the lower Matuyama chron. However, the data from the Bhaun section indicate that rotation in this region was complete by the Gilbert chron. This difference in the time of rotation indicates that the decollement is not rigid but is segmented, with slightly different times of rotation and deformation in different regions.

It has previously been suggested by Crawford (1974) and Klootwijk (1979) that paleomagnetic pole positions from Cambrian and Permian rocks of the Salt Range have suffered rotation of from 30° to 60° in a counter-clockwise direction. This rotation would account for the aberrant position of these paleomagnetic pole positions with respect to the paleozoic polar wander curve for the Indian Plate and in Gondwanaland reconstructions. The results presented here support the hypothesis that these rocks have been rotated counter-clockwise. However, the present study shows that the amount of rotation of the older rocks cannot be uniquely determined due to the apparent fragmental nature of the distal edge of the Salt Range Decollement. In order to accomplish this, systematic paleomagnetic studies will have to be undertaken of Lower Cenozoic and Mesozoic rocks which crop out in the Salt Range. Until this is done, the paleomagnetic results from the Salt Range should not be utilized in plate reconstructions except for those cases in which paleolatitude data are of value.

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